SALESTORM

High-Scale E-Commerce Sale

Stage 7 – API & Event Design

1) Define API method, endpoint, request, response, status, authentication and error handling.

API	Method	Purpose
/api/products/{id}	GET	Get product details
/api/sale/{id}/reserve	POST	Reserve stock
/api/reservations/{id}/confirm	POST	Confirm reservation
/api/reservations/{id}/release	POST	Release stock
/api/payments	POST	Process payment
/api/orders	POST	Create order
/api/orders/{id}	GET	Get order status

API Structure

Client

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API Gateway

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Authentication

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Service

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Business Logic

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Database / Message Queue

Authentication: JWT / OAuth 2.0
Status: 200, 201, 400, 401, 409, 429, 500
Error: Standard error code + message + request ID.

Critical Reservation API

POST /api/sale/{saleId}/reserve

Request

```
{
  "productId": "P100",
  "quantity": 1,
  "idempotencyKey": "USR123-REQ456"
}
```

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Inventory Service

↓

Check Stock → Reserve → Save Reservation

↓

Response

```
{
  "reservationId": "RES789",
  "status": "RESERVED",
  "expiresAt": "..."
}
```

2) Identify commands, events and asynchronous messages.

Commands — "Do this"

ReserveStock

ConfirmReservation

ReleaseReservation

ProcessPayment

CreateOrder

Events — "This happened"

ReservationCreated

PaymentSucceeded

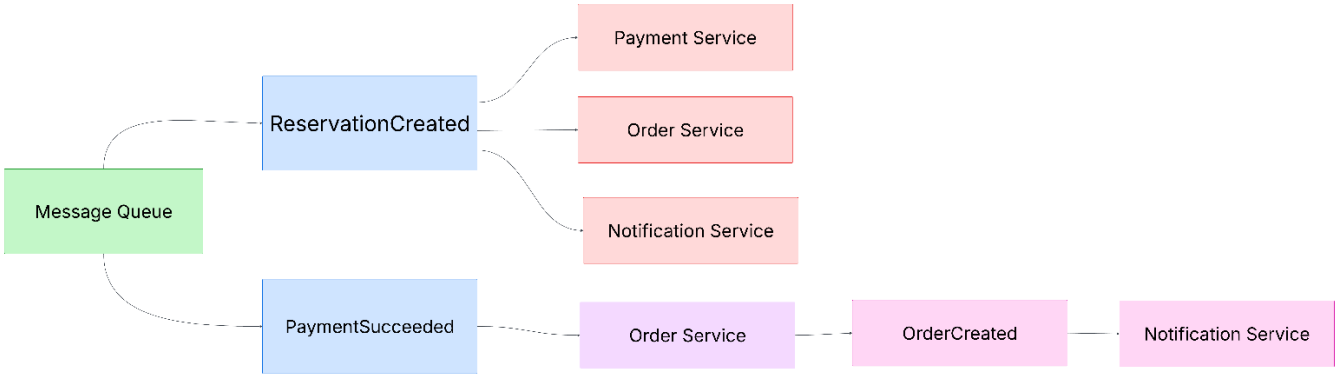
PaymentFailed

ReservationExpired

OrderCreated

OrderFailed

Asynchronous Event Flow



Use idempotency where repeated requests could cause duplicate business transactions.

Idempotency

Idempotency means a repeated request produces the same business result, instead of creating duplicate transactions.

How it works

Each business-changing request contains:

- Idempotency-Key: ABC123

Define event ownership and consumer responsibilities.

Event	Owner / Producer	Consumer	Responsibility
ReservationCreated	Reservation Service	Payment Service	Start payment
PaymentSucceeded	Payment Service	Reservation + Order	Confirm reservation and create order
PaymentFailed	Payment Service	Reservation Service	Release reserved stock
ReservationExpired	Reservation Service	Inventory Service	Return stock
OrderCreated	Order Service	Shipment + Notification	Start fulfillment and notify user
OrderFailed	Order Service	Reservation Service	Trigger compensation/release